



Title of Lecture: Principle of Radiotherapy

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Introduction

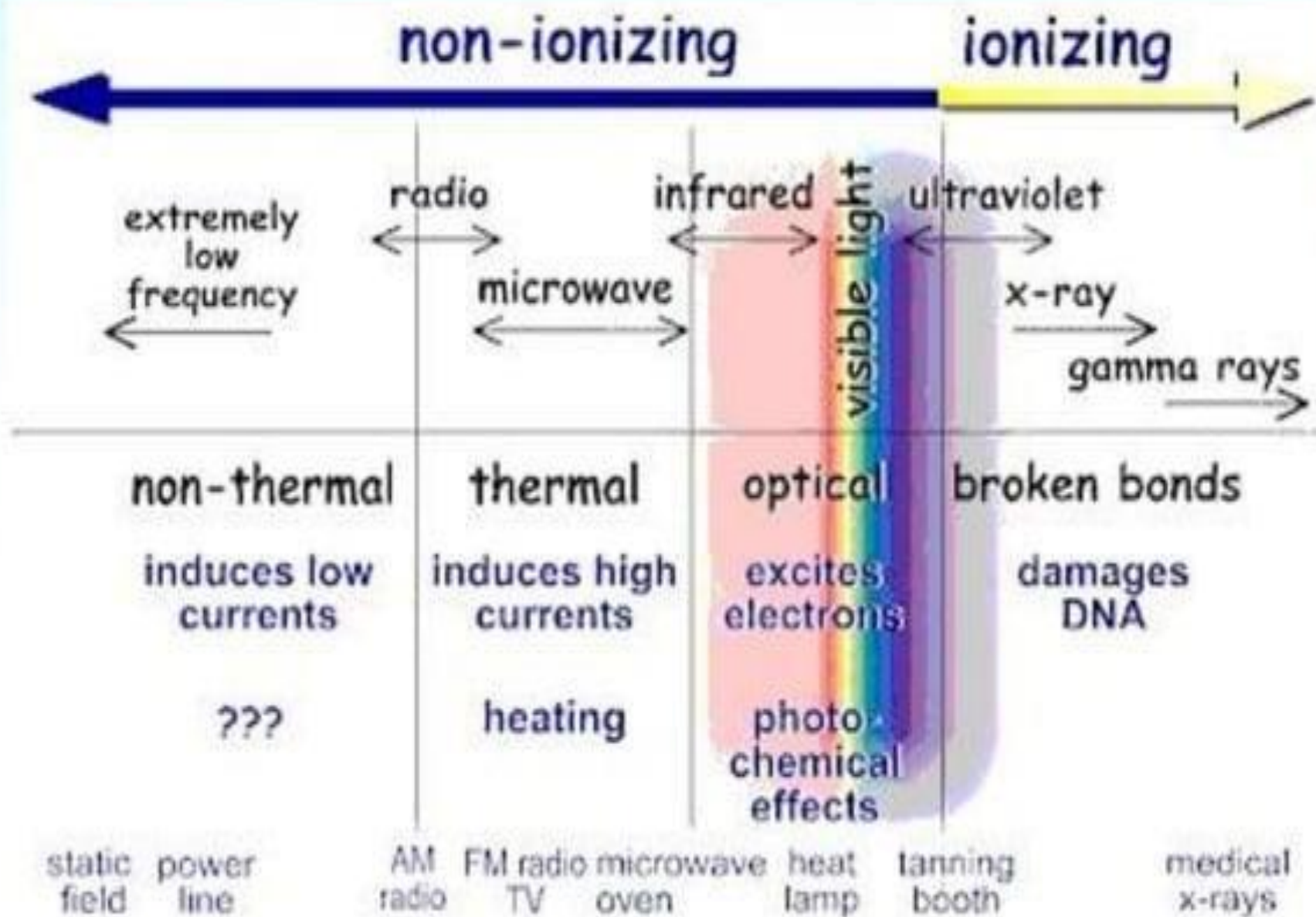
Terminology

- **Radiation oncology** is a specialty focused primarily on the treatment of malignancies
- **Radiation therapists** are the individuals who actually operate the radiation equipment and deliver radiation treatments
- **Dosimetry** - is the discipline of calculating the radiation dose absorbed by the patient and the calculations are based on the depth dose measurements of the radiation beams used to treat an actual patient
- **Radiation physicists** - supervise and review the work of dosimetrists
- **Radiation Unit**
 - Gray (Gy) - The current Standard International unit for absorbed dose
 - One Gy = 100 rad = 1 joule/kg
 - Clinically, the radiation doses for
 - Curative treatment = 70 to 85 Gy
 - Palliative treatment = 30 to 40 Gy

Radiation physics

Radiation

- Energy emitted from a source that is transmitted through an intervening medium or space and absorbed by another body
- Transmission is in the form of waves but wave/particle duality under quantum physics
- Classified as
 - Non-ionizing
 - EM radiation at or below the UV spectrum is nonionizing
 - longer wavelength/lower frequency lower energy
 - May cause injury to humans but the injury is generally limited to thermal damage i.e. burns.
 - Ionizing
 - short wavelength/high frequency higher energy
 - has sufficient energy to produce ions in matter at the molecular level
 - can result significant damage to DNA and denaturation of proteins



- Radiation therapy is the focused delivery of energy in tissue to accomplish controlled biologic damage
- Ionizing radiation used for therapy may be
 - **Electromagnetic radiation**
 - **Particulate radiation**

1) Electromagnetic radiation

- defined by their wave lengths

Photons (x-rays)

- used in external beam therapy
- emitted by the electrons outside the nucleus
 - produced when a stream of electrons collides with a high atomic number target (**tungsten**) located in the head of a linear accelerator

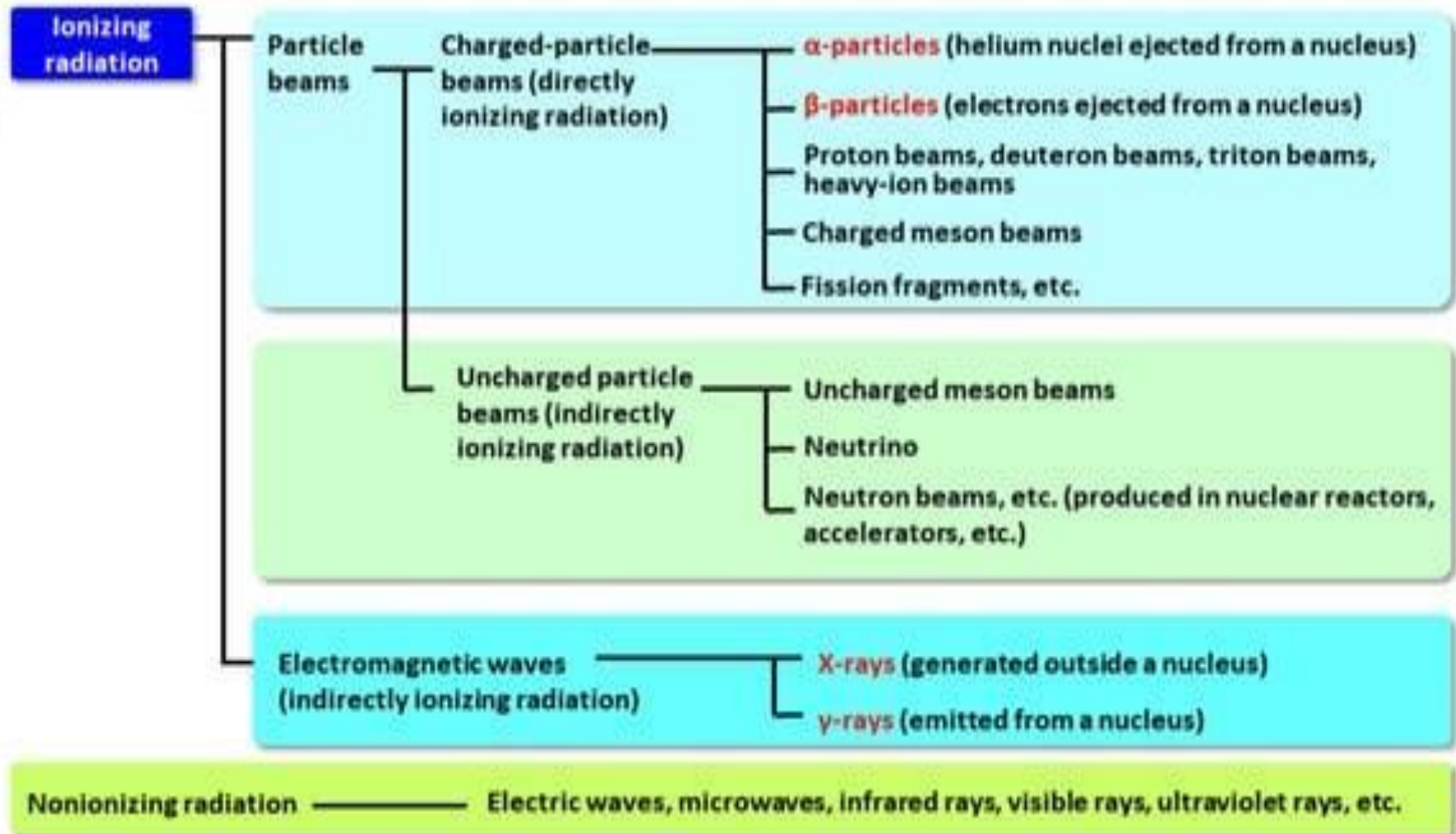
Gamma rays

- emitted by the excited nucleus itself
 - originate from unstable atom nuclei and are emitted during decay of **radioactive** materials (**radionuclides**) which are widely used in **brachytherapy**

2) Particulate radiation

- defined by their masses
- produced by linear accelerators or other high energy generators and are usually delivered by external beam
- electrons, neutrons, protons, helium ions, heavy charged ions, and pi mesons
 - Except for electrons (β -rays), which are available in all modern radiation oncology centers, and protons, other particles have limited clinical use
- **Electron beams:** negatively charged and deposit most of their energy near the surface (used to treat relatively superficial targets)
- **Proton beams:** positively charged particles that are much heavier than electrons
 - Protons scatter minimally as they interact with matter, deposit increasing amounts of energy as they slow down, and then stop at a depth related to their initial energy (**Bragg peak effect**)
 - So they are used for deep targets → **deep pelvic** and **para-aortic lymph nodes**, while sparing unnecessary dosing and injury to anterolateral organs, such as the bowel and kidneys
 - major advantage of proton therapy is the lack of an exit dose through normal tissues. Proton therapy use for gynecologic malignancies is primarily investigational
- **Neutron beams:** neutral particles that tend to deposit most of their energy in a single intranuclear event

Types of Radiation



While radiation includes ionizing radiation and nonionizing radiation, radiation usually means ionizing radiation.

Partially revised "Ionizing Radiation" in the Encyclopedia for Public Acceptance of Atomic Energy Accessible on the Internet, ATOMICA

Therapeutic radiation delivery

Teletherapy

Brachytherapy

Nuclear oncology



- Radiation delivery system
 - Gamma Knife
 - Linac (linear accelerator)
 - CyberKnife
 - Novalis ExacTrac Robotic system
 - Tomotherapy (aka helical tomotherapy)

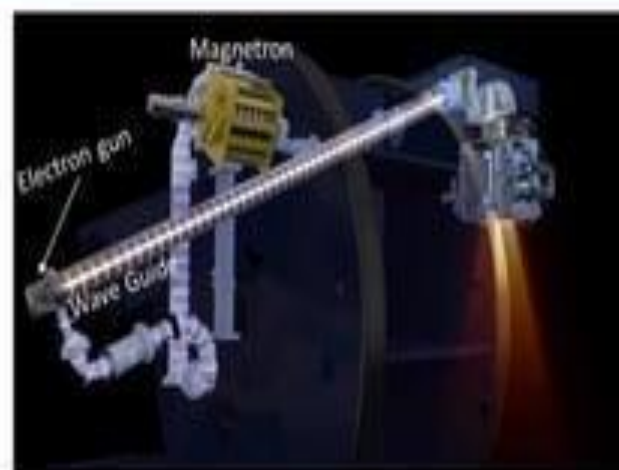
Radiation source

- Radionuclides (radioisotopes)
 - undergo nuclear decay and can emit:
 - (1) positively charged alpha particles,
 - (2) negatively charged beta particles (electrons), and
 - (3) gamma rays
- Linear Accelerator (Linac)
 - can produce both photon and electron beams
 - Photon-therapy mode: for deep-seated tumors
 - Electron-therapy mode: for superficial lesions
 - The unit for electron beam energy is MeV (million electron volts)
- Cobalt machine
 - Uses cobalt-60
 - Produces 1.33 MeV gamma rays
 - short half life, require replacement every 4 to 5 yrs

Physical Properties and Clinical Use of Selected Radionuclides

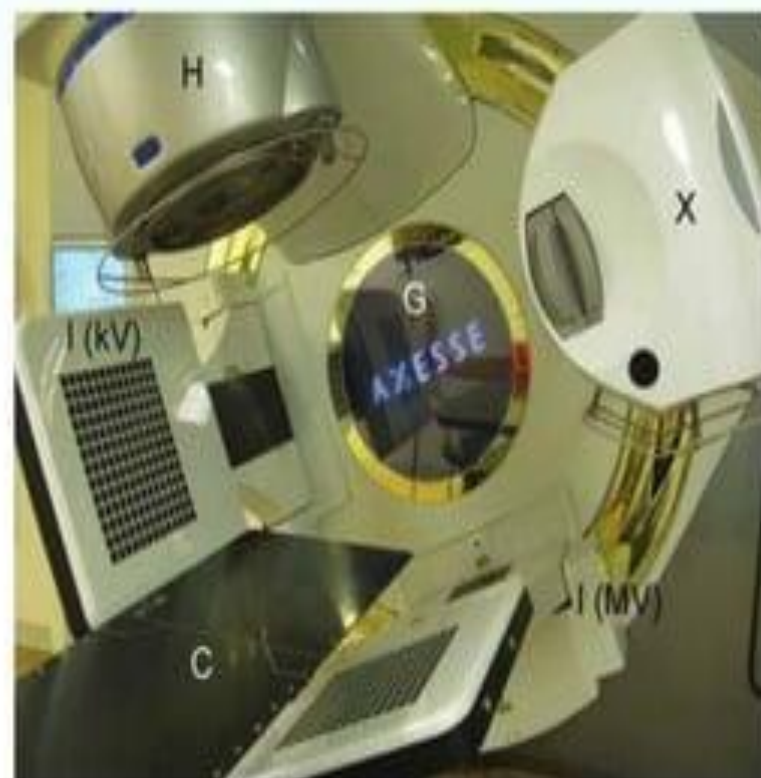
Element	Radiation Energy (MeV)	Half-Life	Clinical Use
Cesium-137	0.6	30 years	Brachytherapy
Iridium-192	0.4	74 days	Brachytherapy
Cobalt-60	1.2	5 years	Brachytherapy
Iodine-125	0.028	60 days	Brachytherapy
Phosphorus-32	1.7	14 days	Intraperitoneal instillation
Gold-196	0.4	2.7 days	Intraperitoneal instillation
Strontium-89	1.4	51 days	Diffuse bone metastasis

MeV= million electron volts.



Linear Accelerator

- The patient lies on the treatment couch (C). The gantry (G), couch, and head (H)
 - These components can all rotate 360 degrees, which allows multiple fields and angles to achieve optimal dose delivery to a tumor
- A kV x-ray source (X) is used in conjunction with an image intensifier (I-kV) for image guided radiotherapy (IGRT)
- Another image intensifier (I-MV) is used for standard port imaging



Linear Accelerator



- ▶ X-rays (also referred to as photons) or electron beams
- ▶ Greater flexibility in treating both deep and superficial tumors
- ▶ High-frequency electromagnetic waves to accelerate charged particles
- ▶ Electron energies - 4 to 30 million electron volts (MeV)
- ▶ source-to-skin (or axis) distance -100 cm

Inverse square law

- radiation intensity from a point source is inversely proportional to the square of the distance away from the radiation source
 - **Dose $\propto (1 \div \text{distance}^2)$**
 - Example: dose at 2 cm will be one-fourth of the dose at 1 cm

Radiation biology

- The exact mechanism of cell death due to radiation is still an area of active investigation

Mechanism of injury	Description
Double strand break of DNA	• DNA injuries involve its strands, bases, and cross-links • Single strand breaks - are easily repaired • the most important lesion is the double strand break → This breakage leads to irreversible loss of the reproductive integrity of the cell and eventual cell death
Ionizing	• Directly ionizing → Fast electrons may directly strike DNA to create damage ◦ seen in particulate radiation • Indirectly ionizing (most common) ◦ contribute to ~ 2/3rd of cell biologic damage ◦ fast electron may interact with water to create a hydroxyl radical, which subsequently interacts with DNA to cause injury

Cell Death after radiation

- two pathways
 - Apoptosis (programmed cell death)
 - aka interphase death
 - Mitotic catastrophe
 - the most common mechanism
 - ← premature or inappropriate entry of cells into mitosis

Four R's of Radiation Biology

- four mechanisms by which cells respond to radiation

1) Repair

- Repair of sublethal damage repair (**SLDR**) and potentially lethal damage repair (**PLDR**)

2) Reassortment

- redistributes surviving cells over cell-cycle phases thus avoiding repeated irradiation in resistant phases
- Cells in mitosis (M) and G2 are most sensitive to radiation

- Conversely, cells in G1 and S (DNA synthesis) phases are less sensitive
- When exposed to radiation, those cells that are in the G2/M phase are killed. During **reassortment**, surviving cell populations restart their progression through the mitotic cycle

3) Repopulation

- The last process seen in SLDR
- tissue's response to replenish the cell pool

4) Reoxygenation

- Cells located within 100 microns of blood capillaries are **oxygenated**, and beyond 100 microns, cells are **hypoxic**
- After radiation, the **oxygenated cells** are killed by chemical intermediates described earlier
- Following cell death, the tumor shrinks and allows hypoxic cells to be positioned within the oxygen diffusion range of blood capillaries and become oxygenated

Radiation oncology practice

- Main issues in radiotherapy
 - Target tumor related
 - **en-bloc**
 - a volume of cancerous tissue
 - its local extensions
 - its regional lymphatics
 - Avoid side effect
 - Minimize normality of tissue quality

Radiation Therapy

Basic steps of radiotherapy

1) **Tissue diagnosis**, stage

2) **Simulation**: Scan to confirm extent of the disease and therapy

- CT scan With IV contrast enhancement - to optimally delineate size and location of all tumor target volumes
- MRI - often useful as an adjunct to the planning CT to help delineate the relevant tumor and normal tissue anatomy
- Fluorodeoxyglucose (FDG) positron emission tomography (PET)/CT
 - To rule out distant metastases, detect involved LNs that were apparently normal on CT alone

3) Contouring: localize the tumor or planning of radiation fields - anatomic areas that will receive a tumoricidal dose (four volumes)

- (1) a gross tumor volume (GTV): encompasses any gross disease
- (2) a clinical target volume (CTV): incorporates any areas at **risk for microscopic tumor spread**;
- (3) a planning target volume (PTV): accounts for uncertainties in treatment planning or delivery such as patient motion or daily set-up error; and
- (4) a volume that defines the normal organs at risk (OAR): will be exposed, albeit to a lesser radiation dose

4) Dose optimization: Prescribe the radiation dose and therapy techniques

- prescribed as definitive, palliative, or adjuvant therapy
- Will it be integrated with surgery and chemotherapy?
- Tool that is particularly helpful in the radiation planning and optimization process
 - dose volume histogram (DVH)
 - computer-generated radiation dose map
 - Intensity-modulated radiation therapy (IMRT)

5) Delivery of Therapy

Combination of RT with surgery & chemotherapy

Radiotherapy + Surgery

- Radiation therapy can be given
 - before, after, or at the time of surgery
- surgical resection and its associated morbidity can often be minimized
- Advantages of preoperative adjuvant radiation
 - decrease the potential for locoregional and distant tumor dissemination and the likelihood of positive surgical margins
 - to avoid extensive surgery in locally advanced cancer (eg – vulvovaginal ca)
 - In unresectable cancers, it can transform them into suitable candidates for a surgical attempt
- Purpose of adjuvant radiation
 - Decrease a high probability for local recurrence if positive margins, lymph node metastases, lymphovascular invasion, and high-grade disease
 - **Cervical ca** with lymphovascular invasion, deep stromal invasion, or large tumor size
 - **Uterine ca**: stage IB or greater disease
 - **Intermediate risk**: older age, lymphovascular invasion, deep myometrial invasion, or intermediate- or high-grade disease
 - In these cases, it is ideally delivered **3 to 6 weeks** following surgery
- Intraoperative radiation therapy (IORT) is infrequent

Normal tissue response to radiation

- Effects of radiation on tissues are generally of two types:
 - Loss of mature functional cells by **apoptosis** (programmed cell death)
 - This usually occurs within 24 hours of radiation
 - Loss of cellular reproductive capacity
 - The severity depends upon the total dose of radiation, length of time over which radiotherapy is delivered and the radiosensitivity of the particular cell types
 - Usually lost cells are replaced by proliferation of surviving stem cells or progenitor cells
- In general, radiation therapy is less well tolerated if
 - (1) the irradiated tissue volume is large,
 - (2) the radiation dose is high,
 - (3) the dose per fraction is large, and
 - (4) the patient's age is advanced

Radiation therapy side effects

Complications of radiotherapy

- Increased incidence of spontaneous mutations
 - the doubling dose is ~100Gy(experiment in mice)
 - 1- 6% of spontaneous mutations in humans
- Permanent sterility
 - 250-600 Gy in acute exposure,0.2 Gy/yr in protracted exposure
- Poor wound healing, fistula formation
- Radiation induced soft tissue necrosis
- Reduced bladder capacity, hematuria, ureteral/urethral stricture
- Hematochezia, bowel obstruction, tenesmus, fecal incontinence, fistula formation
- Dryness and shortening of vagina, dyspareunia, decreased libido

	Short-term side effects	Long-term or late side effects
Fatigue	✓	
Skin problems	✓	
Appetite loss	✓	
Nausea	✓	
Mouth and throat problems	✓	
Bladder problems	✓	✓
Bowel problems	✓	✓
Hair loss	✓	✓
Lymphoedema	✓	✓
Tissue hardening (fibrosis)		✓
Sexuality and intimacy issues	✓	✓
Infertility		✓

Goals of Clinical Radiation therapy

RADICAL /CURATIVE THERAPY

- ❖ Decrease the number of tumor cells to a level that achieves permanent local tumor control

PALLIATIVE RADIOTHERAPY

- ❖ Decrease the symptoms of malignant disease
- ❖ Includes *all or only part* of demonstrated tumor

Staging of tumor is necessary to decide goal of therapy



Thank You